THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The coefficient of x in $3x^2 + 7x + 2$ is: 1

(a) 10 **(b)** 7 **(c)** 3 **(d)** 2

2. If 4 books cost ₹ 120, then 7 books cost: 1

(a) ₹ 214 **(b)** ₹ 127 **(c)** ₹ 210 **(d)** ₹ 180

3. $\frac{3}{5}$ expressed as a percentage is: 1

(a) 60% **(b)** 120% **(c)** 66% **(d)** 54%

4. The legs of a right triangle are 6 cm and 8 cm. Its hypotenuse (in cm) is: 1

(a) 11 **(b)** 14 **(c)** 9 **(d)** 10

5. The ratio 15:25 in its simplest form is: 1

(a) 3:6 **(b)** 4:5 **(c)** 5:3 **(d)** 3:5

6. The value of 101×24 is: 1

(a) 2400 **(b)** 2448 **(c)** 2424 **(d)** 2434

7. The ratio 20 : 35 in simplest form is: 1

(a) 5 : 7 **(b)** 7 : 4 **(c)** 4 : 7 **(d)** 4 : 8

8. 35% written as a fraction in lowest terms is: 1

(a) $\frac{7}{2}$ **(b)** $\frac{40}{100}$ **(c)** $\frac{7}{20}$ **(d)** $\frac{100}{35}$

9. The hypotenuse of a right triangle is 10 cm and one leg is 6 cm. The other leg (in cm) is: 1

(a) 9 **(b)** 8 **(c)** 4 **(d)** 7

10. If 3:5 = 18:x, then x is: 1

(a) 23 **(b)** 30 **(c)** 11 **(d)** 36

11. The value of $6 \times (50 + 4)$ is: 1

(a) 324 **(b)** 330 **(c)** 1200 **(d)** 304

12. If 3 : 5 = 18 : x, then x is: 1

(a) 30 **(b)** 32 **(c)** 24 **(d)** 36

13. 20% of 450 is: 1

(a) 81 **(b)** 180 **(c)** 99 **(d)** 90

14. Which of the following is a Pythagorean triplet? 1

(a) 6, 9, 10 **(b)** 6, 8, 10 **(c)** 6, 8, 11 **(d)** 7, 9, 10

15. ₹ 360 is divided between two friends in the ratio 4:5. The larger share (in ₹) is: 1

(a) 200 **(b)** 220 **(c)** 160 **(d)** 180

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find 34^2 using $(a+b)^2$. 2

17. If 7 notebooks cost ₹ 91, find the cost of 12 notebooks. 2

18. An article is bought for ₹ 400 and sold for ₹ 460. Find the profit. 2

19. Is a triangle with sides 6, 8, 10 cm right-angled? 2

20. Divide ₹ 840 between A and B in the ratio 3:4. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In a pie chart of 200 students' favourite games, cricket takes up 30% of the chart. Find the central angle of the cricket sector and the number of students who like cricket. 3

22. Expand and simplify $3(x+2)+5(x+1)$. 3

23. Divide ₹ 840 between Ria and Sia in the ratio 2 : 5. 3

24. An article costing ₹ 500 is sold for ₹ 450. Find the loss percent. 3

OR

A ladder 10 m long leans against a wall. Its foot is 6 m from the wall. How high on the wall does it reach? 3

25. The scale of a map is 1 cm = 30 km. Two cities are 4 cm apart on the map. Find the actual distance. Also find the map distance of two towns that are actually 150 km apart. 3

OR

Expand $(x+4)(x+5)$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Using $(2m,\ m^2-1,\ m^2+1)$ with $m=2$, write a Pythagorean triplet and verify it. 5

OR

A camp has food for 20 people for 15 days. If 10 more people join the camp, for how many days will the same food last? 5

27. Expand $(x+3)(x+5)$ and find its value when $x=2$. 5

OR

Two numbers are in the ratio 2 : 7 and their sum is 72. Find the numbers and their difference. 5

END OF QUESTION PAPER — **Total: 50 Marks**